

Sergey Gendel

Senior Machine Learning Engineer

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 [LinkedIn Profile](#) |  [GitHub Portfolio](#)

Professional Summary

Senior Machine Learning Engineer with **12+ years of international experience** delivering innovative AI and ML solutions across healthcare and semiconductor metrology. Proven track record of deploying high-impact models, advising on cutting-edge LLM-driven applications, and leading the full ML lifecycle from research to production.

Areas of Expertise

Programming Languages:

- Python (NumPy, Pandas, Scikit-learn, XGBoost), MATLAB, R

ML Frameworks:

- PyTorch, TensorFlow/Keras, Hugging Face Transformers

ML & AI Specializations:

- Classical ML (Regression, Classification, Clustering)
- Deep Learning (CNN, RNN, LSTM)
- Reinforcement Learning (Q-Learning, Deep RL)
- LLM Technologies (RAG, LoRA, LangChain)
- Generative Models (GAN, VAE)

Web & API Development:

- FastAPI, Flask

Databases & Storage:

- MySQL, MongoDB, AWS S3

Cloud Computing & MLOps:

- AWS (SageMaker, Lambda, EC2, API Gateway)

- Docker (Containerization)
- CI/CD (GitHub Actions, AWS CodePipeline)
- Monitoring & Logging (CloudWatch, Weights & Biases)

Collaboration & Management Tools:

- Git, JIRA, Confluence

Languages:

- English (Professional Fluency)
- Hebrew (Native)
- Russian (Native)

Professional Experience

Founder & Principal AI Consultant

GENDEL.AI SERVICES, Ottawa, ON | Jul 2024 – Present

- Consult on the development and deployment of customized Large Language Models (LLMs) and LLM-powered agents utilizing RAG, LangChain, and LoRA fine-tuning.
- Advise on best practices for scalable AI solutions, cloud architecture (AWS), automated CI/CD pipelines, model deployment, orchestration, and monitoring strategies.
- Provide strategic guidance on infrastructure optimization and secure API development.

Senior Machine Learning Engineer

Samsung R&D Center, Israel | Nov 2018 – Jul 2024

- Developed and deployed automated ML systems, achieving up to 50% error reduction in flash memory devices.
- Researched and implemented innovative ML methods to significantly extend flash memory lifetime.
- Collaborated closely with hardware teams to seamlessly integrate AI-driven solutions into Samsung's product lines, substantially enhancing reliability and performance.

Algorithm Specialist

Nova Measuring Instruments Ltd., Israel | Feb 2016 – Oct 2018

- Designed and implemented advanced ML algorithms for precision metrology in semiconductor manufacturing.
- Developed automated systems improving semiconductor production process accuracy and efficiency.
- Conducted research on statistical and mathematical modeling, enhancing existing analytical methods.

AI Engineer

Optimata Ltd., Israel | Feb 2014 – Jan 2016

- Developed biomathematical ML models predicting personalized oncology therapy outcomes.
- Created robust algorithms for predicting cancer progression, significantly aiding medical decision-making.
- Coordinated closely with healthcare professionals, ensuring accuracy, effectiveness, and clinical relevance.

Machine Learning Consultant

Gili Medical Ltd., Israel | Dec 2015 – May 2016

- Engineered predictive ML algorithms to forecast blood sugar fluctuations and detect early hypoglycemia risks using advanced data-driven methods (SVM, Decision Trees, CNN, LSTM, RNN).
- Enhanced prediction accuracy, contributing to the development of life-saving non-invasive medical technology.

2004 – 2012

Teaching Assistant • Technion

Israel Institute of Technology

- Delivered tutorials and graded coursework in Differential Equations and Numerical Analysis.
- Provided individual mentorship, enhancing student comprehension and academic performance.

Education

2014

Technion -The Israel Institute of Technology

Ph.D. in Mechanical Engineering

2007

Technion -The Israel Institute of Technology

M.Sc. in Applied Mathematics

2000

Technion -The Israel Institute of Technology

B.Sc. in Applied Mathematics

Patents

DE-NOISING USING MULTIPLE THRESHOLD-EXPERT MACHINE LEARNING MODELS

<https://patents.justia.com/patent/20220293192>

ERROR DETECTION AND CORRECTION USING MACHINE LEARNING

<https://patents.justia.com/patent/20220013189>

Publications

Kleiman, A., Gendel, S., Degani, D., & Gottlieb, O. (2016). Self-excited dynamics of an elastically restrained slender rigid body in uniform compressible laminar flow. *Nonlinear Dynamics*, 86(4), 2207–2230

Gendel, S., Gottlieb, O., & Degani, D. (2015). Fluid–Structure Interaction of an Elastically Mounted Slender Body at High Incidence. *AIAA Journal*, 53(5), 1309–1318.

Gendel, S., Gottlieb, O., & Degani, D. (2013, June). Flow around an elastically mounted slender body at high incidence. *21st AIAA Computational Fluid Dynamics Conference*.

Gendel, S., Gottlieb, O. & Degani, D. (2012) Nonlinear spatiotemporal fluid-structure interaction of an elastically mounted slender body at high incidence, *Proc. 33rd IACMM Symposium on Computational Mechanics*

Gendel, S., Degani, D. & Gottlieb O. (2010, June) Numerical analysis of the influence of finite rotations on vortex-induced vibrations of a tethered cylinder undergoing internal resonances, *Proc. 6th IUTAM Symp. on Bluff Body Wakes and Vortex-induced Vibrations*, BBVIV-6, 329-332.